```
# OSL PRACTICAL 4
# File Permissions and Ownership in Linux

# 1. Create project directory and files
cd ~/OSL_Practicals
mkdir -p AIML_Project
cd AIML_Project
touch students.txt faculty.txt administrators.txt
# Add content
echo "Student records: Rahul, Priya, Amit" > students.txt
echo "Faculty records: Prof. Sharma, Prof. Patil" > faculty.txt
echo "Administrator records: Admin1, Admin2" > administrators.txt

# 2. Assign different numeric permissions
chmod 644 students.txt
chmod 640 faculty.txt
chmod 600 administrators.txt

# 3. Display ownership and permissions
whoami
id
ls -l
stat students.txt
stat faculty.txt
stat administrators.txt
# 4. Change permissions using symbolic notation
chmod u+x students.txt
chmod g+w faculty.txt
chmod o+r administrators.txt
ls -l

# Change permissions using numeric notation
chmod 755 students.txt
chmod 644 faculty.txt
chmod 700 administrators.txt
ls -l

# 5A. Demonstrate read permission
cat students.txt
chmod 000 students.txt
cat students.txt
chmod 644 students.txt

# 5B. Demonstrate write permission
echo "New student: Karan" >> students.txt
cat students.txt
chmod 444 students.txt
echo "New student: Neha" >> students.txt
chmod 644 students.txt

# 5C. Demonstrate execute permission
echo 'echo "AIML Project Executed Successfully"' > test.sh
chmod 644 test.sh
./test.sh
chmod 755 test.sh
./test.sh
# Final permission details
ls -l
```